Poojan Brahmbhatt

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Education

University of Southern California — Los Angeles, CA
M.S. in Applied Data Science; GPA: 3.5/4.0 — *Jan 2026 – Dec 2027*
- **Coursework:** Machine Learning for Data Science, Data Management, Programming for Data Science

CHARUSAT University — Gujarat, India
B.Tech in Computer Science and Engineering; CGPA: 8.02/10 — *2022 – 2025*
- **Coursework:** Data Structures & Algorithms, Machine Learning, NLP, Modern AI, Operating Systems

Experience

Digikentro — India
AI/ML Intern — *Aug 2025 – Dec 2025*
- Deployed **predictive ML models** and **AI-driven automation** into an **enterprise client app** with a **10+ member** team
- Built end-to-end **ML pipelines** in **Python** (Pandas, scikit-learn) on **10K+ records**, from **feature engineering** to tuning

Charusat Space Research & Technology Center — India
Research Intern — *Dec 2024 – May 2025*
- Benchmarked **U-Net, DeepLabV3+, TransUNet** segmentation on **5,108 satellite images**; best accuracy **83.63%**
- Built the **TensorFlow/Keras** vision pipeline (augmentation, **Dice/IoU** evaluation) and co-authored a manuscript

iNeuron.ai — Remote
Data Science Intern — *May 2024 – Jun 2024*
- **Phishing Detection:** Built **84 URL features** over **58K+ domains**; **Random Forest** reached **94% accuracy/F1**
- **Customer Personality Analysis:** Segmented customers into buyer personas with **K-Means clustering** and EDA

Projects

VeriSearch – Multi-Agent Research Assistant | *Python, LangGraph, Groq LLMs, Tavily, Streamlit* — Live Demo | GitHub
- Built a **4-agent LangGraph system** (Planner, Researcher, Writer, Critic) turning **12 web sources** into cited reports
- Designed an **LLM-as-judge** Critic with **rule-based citation checks**, triggering **self-correcting revisions** in 3/3 test runs

ReviewLens – AI Product Review Analyzer | *Python, scikit-learn, Groq LLMs, SQL, Streamlit* — Live Demo | GitHub
- Trained a **TF-IDF + Logistic Regression** model on **42K reviews**: **94% within one star**, 88–90% on unseen categories
- Used an **LLM for aspect-based sentiment** with **verified quotes**, plus **SQL window functions** for rating analytics

HitScope – Spotify & Billboard Music Trends | *Python, Pandas, scikit-learn, REST API, pytest* — Live Demo | GitHub
- Merged Spotify data, **Spotify Web API**, and **330K Billboard entries**: **75%** of hits charted, **56%** of those top-10
- Applied **K-Means** and **PCA** to find 3 song types and that hits got **57s shorter**; deployed a dashboard with **52 tests**

Publications

A Lightweight Deep Learning Framework for Land Cover Classification from Sentinel-2 Imagery — IEEE Xplore
- 2nd author, **IEEE INCET 2025**: fine-tuned **DenseNet121** on **27K satellite images**, reaching **90.79% accuracy**

Semantic Resume Retrieval for Intelligent Recruitment: A Survey and Experimental Evaluation
- 2nd author, **ICACS 2026** (accepted): **FAISS** semantic search over **2,483 resumes**; MiniLM led with **0.877 NDCG@5**

Technical Skills

Languages: Python, SQL, Java, C/C++
ML & Deep Learning: scikit-learn, PyTorch, TensorFlow/Keras, CNNs, Transfer Learning, Computer Vision, K-Means, PCA
LLMs & NLP: LangGraph, LangChain, RAG, Multi-Agent Systems, FAISS, Sentence Transformers
Data & Tools: Pandas, NumPy, Matplotlib, MySQL, MongoDB, Git, Streamlit, pytest
Certifications: Machine Learning (Simplilearn, 2024), Generative AI (CHARUSAT, 2024)